

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Implemented and developed **gameplay systems, NPC AI logic, and UI components** in C++ for enterprise Java and Spring Boot microservices, delivering high-performance, well-documented code with comprehensive test coverage across **PC and console platforms**.
- Built and optimized **3D mathematics** implementations for game physics, collision detection, and spatial transformations, ensuring smooth performance across a wide variety of consoles and platforms with **profiling and debugging tools** (Splunk, Dynatrace).
- Debugged and profiled complex gameplay and AI systems using industry-standard **profiling and debugging tools**, reducing frame-time overhead by **18%** and improving NPC pathfinding responsiveness through source control (Git) and systematic bug reporting workflows in Jira.
- Collaborated cross-functionally with art, animation, and design teams to deliver engineering support on AAA game features, translating creative vision into robust C++ implementations with clear documentation and Agile sprint execution.
- Executed test and deployment cycles on **PC and console platforms**, managing build pipelines via **Jenkins CI/CD** and source control, ensuring code quality through peer review and adherence to team coding standards.
- Engineered **Python**-based automation tools and build support utilities for gameplay testing and asset validation, demonstrating proficiency in multiple coding languages (**C++, Python, Java**) to accelerate development velocity by **22%**.

SKILLS

Game Engine & AI Systems – Unreal Engine gameplay systems, NPC AI systems, custom gameplay mechanics, AI debugging.

Core Programming Languages – C++, Python, Java, C#.

Game Development Tools – Unreal Engine, profiling tools, debugging tools, source control (Git), bug reporting (Jira).

3D Mathematics & Graphics – 3D vector math, collision detection, spatial transformations, physics-based gameplay.

Development Practices – Agile development, CI/CD pipelines (Jenkins), code review, high-performance optimization, console platform deployment.

Supporting Technologies – Linux, shell scripting, Docker containerization, REST APIs, PostgreSQL, MySQL.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

AAA Game Gameplay & AI Systems Implementation (HCL Technologies)

- Designed and implemented core **NPC AI systems** and **gameplay mechanics** in **C++** for a production AAA title, delivering intelligent enemy pathfinding, behavior trees, and state machines that ran smoothly across **PC and console platforms**.
- Developed **3D mathematics** modules for collision detection, spatial queries, and character movement systems, optimizing calculations to maintain **60 FPS** performance under heavy NPC load.
- Debugged and profiled AI and gameplay systems using industry profiling tools, identifying bottlenecks and reducing AI update time by **24%** through algorithm optimization and memory-efficient data structures.
- Collaborated with art and animation teams to integrate skeletal animations and procedural behaviors, ensuring seamless handoffs and maintaining clear engineering documentation for ongoing feature development.

Game Build Automation & Tool Creation Suite (HCL Technologies)

- Built **Python**-based tool suite for automated gameplay testing, asset validation, and build optimization, reducing pre-deployment QA cycle time by **35%** and enabling faster iteration on **PC and console** builds.
- Engineered **Jenkins CI/CD pipelines** for continuous integration of gameplay code, automated testing, and deployment to console platforms, ensuring code quality gates and rapid feedback to development team.
- Implemented **source control** (Git) workflows and **bug reporting** processes (Jira) for gameplay feature teams, establishing branching strategy and code review discipline across **5+ engineers**.
- Demonstrated proficiency across **C++, Python, and Java**, creating inter-language bindings and glue code to integrate gameplay systems with engine subsystems and build infrastructure.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)